TUGAS PENDAHULUAN 3

Nama : Nadia Tambunan
NIM : 103122400005
Kelas : SE-08-01

1. Buat Tabel

Pertemuan1.sql | Welcome Page | TP3

Worksheet | Query Builder

```
CREATE TABLE mahasiswa (
id char(12),
nama VARCHAR(100),
tempat_lahir VARCHAR(100),
tanggal_lahir DATE,
nomor_hp char(14),
email VARCHAR(100),
tinggi_badan INT,
berat_badan INT);
```

Script Output | Task completed in 0.218 seconds

```
nomor_hp char(14),
email VARCHAR(100),
tinggi_badan INT PRIMARY KEY,
berat_badan INT PRIMARY KEY)
Error report -
ORA-02260: table can have only one primary key

https://docs.oracle.com/error-help/db/ora-02260/02260. 00000 -  "table can have only one primary key"
*Cause:    Self-evident.
*Action:   Remove the extra primary key.

Table MAHASISWA created.
```

```
SELECT * FROM mahasiswa;
```

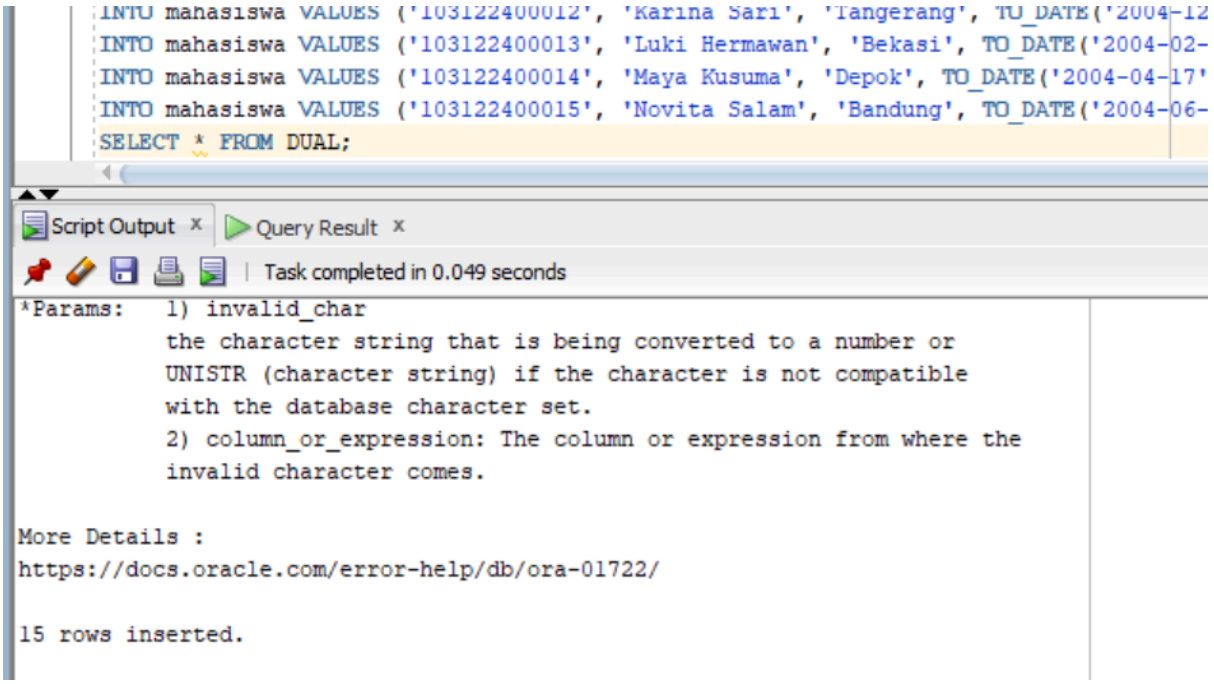
Script Output | Query Result

SQL | All Rows Fetched: 0 in 0.029 seconds

ID	NAMA	TEMPAT_...	TANGGAL...	NOMOR_HP	EMAIL	TINGGI_B...	BERAT_B...

2. Insert 15 Rows (data saya minta AI untuk buatkan)

```
INSERT ALL
INTO mahasiswa VALUES ('103122400001', 'Ahmad Wijaya', 'Jakarta', TO_DATE('2004-01-15', 'YYYY-MM-DD'),
'0812345678901', 'ahmad@gmail.com', 175, 72)
INTO mahasiswa VALUES ('103122400002', 'Budi Santoso', 'Bandung', TO_DATE('2004-03-22', 'YYYY-MM-DD'),
'0821234567890', 'budi@gmail.com', 172, 68)
INTO mahasiswa VALUES ('103122400003', 'Citra Dewi', 'Surabaya', TO_DATE('2004-05-10', 'YYYY-MM-DD'),
'0831234567890', 'citra@gmail.com', 162, 55)
INTO mahasiswa VALUES ('103122400004', 'Doni Prasetyo', 'Medan', TO_DATE('2004-07-18', 'YYYY-MM-DD'),
'0841234567890', 'doni@gmail.com', 178, 75)
INTO mahasiswa VALUES ('103122400005', 'Nadia Tambunan', 'Jakarta', TO_DATE('2004-02-06', 'YYYY-MM-DD'),
'0882008729098', 'nadiatambunan@gmail.com', 164, 61)
INTO mahasiswa VALUES ('103122400006', 'Eka Putra', 'Yogyakarta', TO_DATE('2004-09-12', 'YYYY-MM-DD'),
'0851234567890', 'eka@gmail.com', 170, 70)
INTO mahasiswa VALUES ('103122400007', 'Fiona Mayang', 'Semarang', TO_DATE('2004-11-28', 'YYYY-MM-DD'),
'0861234567890', 'fiona@gmail.com', 165, 58)
INTO mahasiswa VALUES ('103122400008', 'Gina Septiani', 'Makassar', TO_DATE('2004-04-05', 'YYYY-MM-DD'),
'0871234567890', 'gina@gmail.com', 160, 52)
INTO mahasiswa VALUES ('103122400009', 'Hendra Gunawan', 'Jakarta', TO_DATE('2004-06-14', 'YYYY-MM-DD'),
'0881234567890', 'hendra@gmail.com', 180, 78)
INTO mahasiswa VALUES ('103122400010', 'Ica Ramadhani', 'Palembang', TO_DATE('2004-08-21', 'YYYY-MM-DD'),
'0891234567890', 'ica@gmail.com', 163, 56)
INTO mahasiswa VALUES ('103122400011', 'Jaka Permana', 'Bogor', TO_DATE('2004-10-30', 'YYYY-MM-DD'),
'0801234567890', 'jaka@gmail.com', 176, 74)
INTO mahasiswa VALUES ('103122400012', 'Karina Sari', 'Tangerang', TO_DATE('2004-12-11', 'YYYY-MM-DD'),
'0811234567890', 'karina@gmail.com', 164, 60)
INTO mahasiswa VALUES ('103122400013', 'Luki Hermawan', 'Bekasi', TO_DATE('2004-02-25', 'YYYY-MM-DD'),
'0821234567891', 'luki@gmail.com', 179, 76)
INTO mahasiswa VALUES ('103122400014', 'Maya Kusuma', 'Depok', TO_DATE('2004-04-17', 'YYYY-MM-DD'),
'0831234567891', 'maya@gmail.com', 161, 54)
INTO mahasiswa VALUES ('103122400015', 'Novita Salam', 'Bandung', TO_DATE('2004-06-03', 'YYYY-MM-DD'),
'0841234567891', 'novita@gmail.com', 166, 62)
SELECT * FROM DUAL;
```



Script Output × Query Result ×

SQL | All Rows Fetched: 15 in 0.021 seconds

	ID	NAMA	TEMPAT_LAHIR	TANGGAL_LAHIR	NOMOR_HP	EMAIL	TINGGI_BADAN	BERAT_BADAN
1	103122400001	Ahmad Wijaya	Jakarta	15-01-2004	0812345678901	ahmad@gmail.com	175	72
2	103122400002	Budi Santoso	Bandung	22-03-2004	0821234567890	budi@gmail.com	172	68
3	103122400003	Citra Dewi	Surabaya	10-05-2004	0831234567890	citra@gmail.com	162	55
4	103122400004	Doni Prasetyo	Medan	18-07-2004	0841234567890	doni@gmail.com	178	75
5	103122400005	Nadia Tambunan	Jakarta	06-02-2004	0882008729098	nadiatambunan@gmail.com	164	61
6	103122400006	Eka Putra	Yogyakarta	12-09-2004	0851234567890	eka@gmail.com	170	70
7	103122400007	Fiona Mayang	Semarang	28-11-2004	0861234567890	fiona@gmail.com	165	58
8	103122400008	Gina Septiani	Makassar	05-04-2004	0871234567890	gina@gmail.com	160	52
9	103122400009	Hendra Gunawan	Jakarta	14-06-2004	0881234567890	hendra@gmail.com	180	78
10	103122400010	Ica Ramadhani	Palembang	21-08-2004	0891234567890	ica@gmail.com	163	56
11	103122400011	Jaka Permana	Bogor	30-10-2004	0801234567890	jaka@gmail.com	176	74
12	103122400012	Karina Sari	Tangerang	11-12-2004	0811234567890	karina@gmail.com	164	60
13	103122400013	Luki Hermawan	Bekasi	25-02-2004	0821234567891	luki@gmail.com	179	76
14	103122400014	Maya Kusuma	Depok	17-04-2004	0831234567891	maya@gmail.com	161	54
15	103122400015	Novita Salam	Bandung	03-06-2004	0841234567891	novita@gmail.com	166	62

3. proses pemrosesan satu table
   a. Fungsi Karakter dan String (LOWER/UPPER, INITCAP, CONCAT, LENGTH, SUBSTR, LPAD/RPAD, LTRIM, RTRIM, INSTR, REPLACE)

```
-- LOWER: Mengubah nama menjadi huruf kecil
SELECT
    id,
    LOWER(nama) AS nama_lowercase,
    nama
FROM mahasiswa;
```

Script Output × Query Result ×

SQL | All Rows Fetched: 15 in 0.011 seconds

	ID	NAMA_LOWERCASE	NAMA
1	103122400001	ahmad wijaya	Ahmad Wijaya
2	103122400002	budi santoso	Budi Santoso
3	103122400003	citra dewi	Citra Dewi
4	103122400004	doni prasetyo	Doni Prasetyo
5	103122400005	nadia tambunan	Nadia Tambunan
6	103122400006	eka putra	Eka Putra
7	103122400007	fiona mayang	Fiona Mayang
8	103122400008	gina septiani	Gina Septiani
9	103122400009	hendra gunawan	Hendra Gunawan
10	103122400010	ica ramadhani	Ica Ramadhani
11	103122400011	jaka permana	Jaka Permana
12	103122400012	karina sari	Karina Sari
13	103122400013	luki hermawan	Luki Hermawan
14	103122400014	maya kusuma	Maya Kusuma
15	103122400015	novita salam	Novita Salam

```sql
-- UPPER: Mengubah nama menjadi huruf besar
SELECT
    id,
    UPPER(nama) AS nama_uppercase,
    nama
FROM mahasiswa;
```

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.006 seconds

	ID	NAMA_UPPERCASE	NAMA
1	103122400001	AHMAD WIJAYA	Ahmad Wijaya
2	103122400002	BUDI SANTOSO	Budi Santoso
3	103122400003	CITRA DEWI	Citra Dewi
4	103122400004	DONI PRASETYO	Doni Prasetyo
5	103122400005	NADIA TAMBUNAN	Nadia Tambunan
6	103122400006	EKA PUTRA	Eka Putra
7	103122400007	FIONA MAYANG	Fiona Mayang
8	103122400008	GINA SEPTIANI	Gina Septiani
9	103122400009	HENDRA GUNAWAN	Hendra Gunawan
10	103122400010	ICA RAMADHANI	Ica Ramadhani
11	103122400011	JAKA PERMANA	Jaka Permana
12	103122400012	KARINA SARI	Karina Sari
13	103122400013	LUKI HERMAWAN	Luki Hermawan
14	103122400014	MAYA KUSUMA	Maya Kusuma
15	103122400015	NOVITA SALAM	Novita Salam

Worksheet | Query Builder

```sql
-- INITCAP: Huruf pertama setiap kata besar
SELECT
    id,
    INITCAP(nama) AS nama_initcap,
    nama
FROM mahasiswa;
```

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.015 seconds

	ID	NAMA_INITCAP	NAMA
1	103122400001	Ahmad Wijaya	Ahmad Wijaya
2	103122400002	Budi Santoso	Budi Santoso
3	103122400003	Citra Dewi	Citra Dewi
4	103122400004	Doni Prasetyo	Doni Prasetyo
5	103122400005	Nadia Tambunan	Nadia Tambunan
6	103122400006	Eka Putra	Eka Putra
7	103122400007	Fiona Mayang	Fiona Mayang
8	103122400008	Gina Septiani	Gina Septiani
9	103122400009	Hendra Gunawan	Hendra Gunawan
10	103122400010	Ica Ramadhani	Ica Ramadhani
11	103122400011	Jaka Permana	Jaka Permana
12	103122400012	Karina Sari	Karina Sari
13	103122400013	Luki Hermawan	Luki Hermawan
14	103122400014	Maya Kusuma	Maya Kusuma
15	103122400015	Novita Salam	Novita Salam

```sql
-- CONCAT: Menggabungkan kolom
SELECT
    id,
    CONCAT(nama, ' dari ') || tempat_lahir AS nama_tempat
FROM mahasiswa;
```

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.008 seconds

	ID	NAMA_TEMPAT
1	103122400001	Ahmad Wijaya dari Jakarta
2	103122400002	Budi Santoso dari Bandung
3	103122400003	Citra Dewi dari Surabaya
4	103122400004	Doni Prasetyo dari Medan
5	103122400005	Nadia Tambunan dari Jakarta
6	103122400006	Eka Putra dari Yogyakarta
7	103122400007	Fiona Mayang dari Semarang
8	103122400008	Gina Septiani dari Makassar
9	103122400009	Hendra Gunawan dari Jakarta
10	103122400010	Ica Ramadhani dari Palembang
11	103122400011	Jaka Permana dari Bogor
12	103122400012	Karina Sari dari Tangerang
13	103122400013	Luki Hermawan dari Bekasi
14	103122400014	Maya Kusuma dari Depok
15	103122400015	Novita Salam dari Bandung

```sql
-- LENGTH: Panjang string nama
SELECT
    id,
    nama,
    LENGTH(nama) AS panjang_nama
FROM mahasiswa;
```

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.006 seconds

	ID	NAMA	PANJANG_NAMA
1	103122400001	Ahmad Wijaya	12
2	103122400002	Budi Santoso	12
3	103122400003	Citra Dewi	10
4	103122400004	Doni Prasetyo	13
5	103122400005	Nadia Tambunan	14
6	103122400006	Eka Putra	9
7	103122400007	Fiona Mayang	12
8	103122400008	Gina Septiani	13
9	103122400009	Hendra Gunawan	14
10	103122400010	Ica Ramadhani	13
11	103122400011	Jaka Permana	12
12	103122400012	Karina Sari	11
13	103122400013	Luki Hermawan	13
14	103122400014	Maya Kusuma	11
15	103122400015	Novita Salam	12

```sql
-- SUBSTR: Mengambil substring dari nama (3 karakter pertama)
SELECT
    id,
    nama,
    SUBSTR(nama, 1, 3) AS nama_singkat
FROM mahasiswa;
```

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.007 seconds

	ID	NAMA	NAMA_SINGKAT
1	103122400001	Ahmad Wijaya	Ahm
2	103122400002	Budi Santoso	Bud
3	103122400003	Citra Dewi	Cit
4	103122400004	Doni Prasetyo	Don
5	103122400005	Nadia Tambunan	Nad
6	103122400006	Eka Putra	Eka
7	103122400007	Fiona Mayang	Fio
8	103122400008	Gina Septiani	Gin
9	103122400009	Hendra Gunawan	Hen
10	103122400010	Ica Ramadhani	Ica
11	103122400011	Jaka Permana	Jak
12	103122400012	Karina Sari	Kar
13	103122400013	Luki Hermawan	Luk
14	103122400014	Maya Kusuma	May
15	103122400015	Novita Salam	Nov

```sql
-- LPAD: Menambah padding di kiri
SELECT
    id,
    LPAD(id, 15, '*') AS id_lpad
FROM mahasiswa;
```

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.013 second

	ID	ID_LPAD
1	103122400001	***103122400001
2	103122400002	***103122400002
3	103122400003	***103122400003
4	103122400004	***103122400004
5	103122400005	***103122400005
6	103122400006	***103122400006
7	103122400007	***103122400007
8	103122400008	***103122400008
9	103122400009	***103122400009
10	103122400010	***103122400010
11	103122400011	***103122400011
12	103122400012	***103122400012
13	103122400013	***103122400013
14	103122400014	***103122400014
15	103122400015	***103122400015

```sql
-- RPAD: Menambah padding di kanan
SELECT
    id,
    RPAD(nama, 20, '.') AS nama_rpad
FROM mahasiswa;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 15 in 0.007 seconds

	ID	NAMA_RPAD
1	103122400001	Ahmad Wijaya.....
2	103122400002	Budi Santoso.....
3	103122400003	Citra Dewi.....
4	103122400004	Doni Prasetyo.....
5	103122400005	Nadia Tambunan.....
6	103122400006	Eka Putra.....
7	103122400007	Fiona Mayang.....
8	103122400008	Gina Septiani.....
9	103122400009	Hendra Gunawan.....
10	103122400010	Ica Ramadhani.....
11	103122400011	Jaka Permana.....
12	103122400012	Karina Sari.....
13	103122400013	Luki Hermawan.....
14	103122400014	Maya Kusuma.....
15	103122400015	Novita Salam.....

```sql
-- LTRIM: Menghapus spasi di kiri
SELECT
    id,
    LTRIM('    ' || nama) AS nama_ltrim
FROM mahasiswa;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 15 in 0.036 seconds

	ID	NAMA_LTRIM
1	103122400001	Ahmad Wijaya
2	103122400002	Budi Santoso
3	103122400003	Citra Dewi
4	103122400004	Doni Prasetyo
5	103122400005	Nadia Tambunan
6	103122400006	Eka Putra
7	103122400007	Fiona Mayang
8	103122400008	Gina Septiani
9	103122400009	Hendra Gunawan
10	103122400010	Ica Ramadhani
11	103122400011	Jaka Permana
12	103122400012	Karina Sari
13	103122400013	Luki Hermawan
14	103122400014	Maya Kusuma
15	103122400015	Novita Salam

```sql
-- RTRIM: Menghapus spasi di kanan
SELECT
    id,
    RTRIM(nama || '    ') AS nama_rtrim
FROM mahasiswa;
```

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.007 seconds

	ID	NAMA_RTRIM
1	103122400001	Ahmad Wijaya
2	103122400002	Budi Santoso
3	103122400003	Citra Dewi
4	103122400004	Doni Prasetyo
5	103122400005	Nadia Tambunan
6	103122400006	Eka Putra
7	103122400007	Fiona Mayang
8	103122400008	Gina Septiani
9	103122400009	Hendra Gunawan
10	103122400010	Ica Ramadhani
11	103122400011	Jaka Permana
12	103122400012	Karina Sari
13	103122400013	Luki Hermawan
14	103122400014	Maya Kusuma
15	103122400015	Novita Salam

```sql
-- INSTR: Mencari posisi karakter dalam string
SELECT
    id,
    email,
    INSTR(email, '@') AS posisi_at
FROM mahasiswa;
```

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.006 seconds

	ID	EMAIL	POSISI_AT
1	103122400001	ahmad@gmail.com	6
2	103122400002	budi@gmail.com	5
3	103122400003	citra@gmail.com	6
4	103122400004	doni@gmail.com	5
5	103122400005	nadiatambunan@gmail.com	14
6	103122400006	eka@gmail.com	4
7	103122400007	fiona@gmail.com	6
8	103122400008	gina@gmail.com	5
9	103122400009	hendra@gmail.com	7
10	103122400010	ica@gmail.com	4
11	103122400011	jaka@gmail.com	5
12	103122400012	karina@gmail.com	7
13	103122400013	luki@gmail.com	5
14	103122400014	maya@gmail.com	5
15	103122400015	novita@gmail.com	7

```sql
-- REPLACE: Mengganti karakter dalam string
SELECT
    id,
    email,
    REPLACE(email, '@gmail.com', '@yahoo.com') AS email_baru
FROM mahasiswa;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 15 in 0.013 seconds

	ID	EMAIL	EMAIL_BARU
1	103122400001	ahmad@gmail.com	ahmad@yahoo.com
2	103122400002	budi@gmail.com	budi@yahoo.com
3	103122400003	citra@gmail.com	citra@yahoo.com
4	103122400004	doni@gmail.com	doni@yahoo.com
5	103122400005	nadiatambunan@gmail.com	nadiatambunan@yahoo.com
6	103122400006	eka@gmail.com	eka@yahoo.com
7	103122400007	fiona@gmail.com	fiona@yahoo.com
8	103122400008	gina@gmail.com	gina@yahoo.com
9	103122400009	hendra@gmail.com	hendra@yahoo.com
10	103122400010	ica@gmail.com	ica@yahoo.com
11	103122400011	jaka@gmail.com	jaka@yahoo.com
12	103122400012	karina@gmail.com	karina@yahoo.com
13	103122400013	luki@gmail.com	luki@yahoo.com
14	103122400014	maya@gmail.com	maya@yahoo.com
15	103122400015	novita@gmail.com	novita@yahoo.com

b. Fungsi Tanggal dan Waktu (MM/MON/MONTH, DD/DY, YYYY/YY/RR, AM (or PM)/HH/HH24, MI/SS, ADD_MONTHS(d,n), SYSDATE, LAST_DAY(d), NEXT_DAY(d,day_name), MONTHS_BETWEEN(d1,d2), ROUND(d,[MONTH| YEAR]))

```sql
-- MM/MON/MONTH: Format bulan
SELECT
    id,
    nama,
    tanggal_lahir,
    TO_CHAR(tanggal_lahir, 'MM') AS bulan_angka,
    TO_CHAR(tanggal_lahir, 'MON') AS bulan_singkat,
    TO_CHAR(tanggal_lahir, 'MONTH') AS bulan_lengkap
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 15 in 0.012 seconds

	ID	NAMA	TANGGAL_LAHIR	BULAN_ANGKA	BULAN_SINGKAT	BULAN_LENGKAP
1	103122400001	Ahmad Wijaya	15-01-2004	01	JAN	JANUARY
2	103122400002	Budi Santoso	22-03-2004	03	MAR	MARCH
3	103122400003	Citra Dewi	10-05-2004	05	MAY	MAY
4	103122400004	Doni Prasetyo	18-07-2004	07	JUL	JULY
5	103122400005	Nadia Tambunan	06-02-2004	02	FEB	FEBRUARY
6	103122400006	Eka Putra	12-09-2004	09	SEP	SEPTEMBER
7	103122400007	Fiona Mayang	28-11-2004	11	NOV	NOVEMBER
8	103122400008	Gina Septiani	05-04-2004	04	APR	APRIL
9	103122400009	Hendra Gunawan	14-06-2004	06	JUN	JUNE
10	103122400010	Ica Ramadhani	21-08-2004	08	AUG	AUGUST
11	103122400011	Jaka Permana	30-10-2004	10	OCT	OCTOBER
12	103122400012	Karina Sari	11-12-2004	12	DEC	DECEMBER
13	103122400013	Luki Hermawan	25-02-2004	02	FEB	FEBRUARY
14	103122400014	Maya Kusuma	17-04-2004	04	APR	APRIL
15	103122400015	Novita Salam	03-06-2004	06	JUN	JUNE

```sql
-- DD/DY: Format hari
SELECT
    id,
    nama,
    tanggal_lahir,
    TO_CHAR(tanggal_lahir, 'DD') AS tanggal,
    TO_CHAR(tanggal_lahir, 'DY') AS hari_singkat,
    TO_CHAR(tanggal_lahir, 'DAY') AS hari_lengkap
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 15 in 0.005 seconds

	ID	NAMA	TANGGAL_LAHIR	TANGGAL	HARI_SINGKAT	HARI_LENGKAP
1	103122400001	Ahmad Wijaya	15-01-2004	15	THU	THURSDAY
2	103122400002	Budi Santoso	22-03-2004	22	MON	MONDAY
3	103122400003	Citra Dewi	10-05-2004	10	MON	MONDAY
4	103122400004	Doni Prasetyo	18-07-2004	18	SUN	SUNDAY
5	103122400005	Nadia Tambunan	06-02-2004	06	FRI	FRIDAY
6	103122400006	Eka Putra	12-09-2004	12	SUN	SUNDAY
7	103122400007	Fiona Mayang	28-11-2004	28	SUN	SUNDAY
8	103122400008	Gina Septiani	05-04-2004	05	MON	MONDAY
9	103122400009	Hendra Gunawan	14-06-2004	14	MON	MONDAY
10	103122400010	Ica Ramadhani	21-08-2004	21	SAT	SATURDAY
11	103122400011	Jaka Permana	30-10-2004	30	SAT	SATURDAY
12	103122400012	Karina Sari	11-12-2004	11	SAT	SATURDAY
13	103122400013	Luki Hermawan	25-02-2004	25	WED	WEDNESDAY
14	103122400014	Maya Kusuma	17-04-2004	17	SAT	SATURDAY
15	103122400015	Novita Salam	03-06-2004	03	THU	THURSDAY

```sql
-- YYYY/YY/RR: Format tahun
SELECT
    id,
    nama,
    tanggal_lahir,
    TO_CHAR(tanggal_lahir, 'YYYY') AS tahun_4digit,
    TO_CHAR(tanggal_lahir, 'YY') AS tahun_2digit,
    TO_CHAR(tanggal_lahir, 'RR') AS tahun_rr
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 15 in 0.004 seconds

	ID	NAMA	TANGGAL_LAHIR	TAHUN_4DIGIT	TAHUN_2DIGIT	TAHUN_RR
1	103122400001	Ahmad Wijaya	15-01-2004	2004	04	04
2	103122400002	Budi Santoso	22-03-2004	2004	04	04
3	103122400003	Citra Dewi	10-05-2004	2004	04	04
4	103122400004	Doni Prasetyo	18-07-2004	2004	04	04
5	103122400005	Nadia Tambunan	06-02-2004	2004	04	04
6	103122400006	Eka Putra	12-09-2004	2004	04	04
7	103122400007	Fiona Mayang	28-11-2004	2004	04	04
8	103122400008	Gina Septiani	05-04-2004	2004	04	04
9	103122400009	Hendra Gunawan	14-06-2004	2004	04	04
10	103122400010	Ica Ramadhani	21-08-2004	2004	04	04
11	103122400011	Jaka Permana	30-10-2004	2004	04	04
12	103122400012	Karina Sari	11-12-2004	2004	04	04
13	103122400013	Luki Hermawan	25-02-2004	2004	04	04
14	103122400014	Maya Kusuma	17-04-2004	2004	04	04
15	103122400015	Novita Salam	03-06-2004	2004	04	04

```sql
-- AM/PM: Format waktu dengan AM/PM
SELECT
    id,
    nama,
    SYSDATE,
    TO_CHAR(SYSDATE, 'HH:MI:SS AM') AS waktu_am_pm,
    TO_CHAR(SYSDATE, 'HH24:MI:SS') AS waktu_24jam
FROM mahasiswa
WHERE ROWNUM <= 3;
```

Script Output | Query Result

All Rows Fetched: 3 in 0.007 seconds

	ID	NAMA	SYSDATE	WAKTU_AM_PM	WAKTU_24JAM
1	103122400001	Ahmad Wijaya	04-03-2026	12:28:54 PM	12:28:54
2	103122400002	Budi Santoso	04-03-2026	12:28:54 PM	12:28:54
3	103122400003	Citra Dewi	04-03-2026	12:28:54 PM	12:28:54

```sql
-- HH24/MI/SS: Format waktu 24 jam
SELECT
    id,
    nama,
    SYSDATE,
    TO_CHAR(SYSDATE, 'HH24:MI:SS') AS waktu_24jam,
    TO_CHAR(SYSDATE, 'YYYY-MM-DD HH24:MI:SS') AS waktu_lengkap
FROM mahasiswa
WHERE ROWNUM <= 3;
```

Script Output | Query Result

All Rows Fetched: 3 in 0.006 seconds

	ID	NAMA	SYSDATE	WAKTU_24JAM	WAKTU_LENGKAP
1	103122400001	Ahmad Wijaya	04-03-2026	12:29:24	2026-03-04 12:29:24
2	103122400002	Budi Santoso	04-03-2026	12:29:24	2026-03-04 12:29:24
3	103122400003	Citra Dewi	04-03-2026	12:29:24	2026-03-04 12:29:24

```sql
-- ADD_MONTHS: Menambah bulan ke tanggal lahir
SELECT
    id,
    nama,
    tanggal_lahir,
    ADD_MONTHS(tanggal_lahir, 6) AS tanggal_plus_6bulan
FROM mahasiswa;
```

Script Output | Query Result

All Rows Fetched: 15 in 0.006 seconds

	ID	NAMA	TANGGAL_LAHIR	TANGGAL_PLUS_6BULAN
1	103122400001	Ahmad Wijaya	15-01-2004	15-07-2004
2	103122400002	Budi Santoso	22-03-2004	22-09-2004
3	103122400003	Citra Dewi	10-05-2004	10-11-2004
4	103122400004	Doni Prasetyo	18-07-2004	18-01-2005
5	103122400005	Nadia Tambunan	06-02-2004	06-08-2004
6	103122400006	Eka Putra	12-09-2004	12-03-2005
7	103122400007	Fiona Mayang	28-11-2004	28-05-2005
8	103122400008	Gina Septiani	05-04-2004	05-10-2004
9	103122400009	Hendra Gunawan	14-06-2004	14-12-2004
10	103122400010	Ica Ramadhani	21-08-2004	21-02-2005
11	103122400011	Jaka Permana	30-10-2004	30-04-2005
12	103122400012	Karina Sari	11-12-2004	11-06-2005
13	103122400013	Luki Hermawan	25-02-2004	25-08-2004
14	103122400014	Maya Kusuma	17-04-2004	17-10-2004
15	103122400015	Novita Salam	03-06-2004	03-12-2004

```sql
-- SYSDATE: Tanggal sistem saat ini
SELECT
    id,
    nama,
    tanggal_lahir,
    SYSDATE AS tanggal_hari_ini
FROM mahasiswa
WHERE ROWNUM <= 5;
```

Script Output | Query Result

SQL | All Rows Fetched: 5 in 0.007 seconds

	ID	NAMA	TANGGAL_LAHIR	TANGGAL_HARI_INI
1	103122400001	Ahmad Wijaya	15-01-2004	04-03-2026
2	103122400002	Budi Santoso	22-03-2004	04-03-2026
3	103122400003	Citra Dewi	10-05-2004	04-03-2026
4	103122400004	Doni Prasetyo	18-07-2004	04-03-2026
5	103122400005	Nadia Tambunan	06-02-2004	04-03-2026

```sql
-- LAST_DAY: Hari terakhir bulan
SELECT
    id,
    nama,
    tanggal_lahir,
    LAST_DAY(tanggal_lahir) AS hari_terakhir_bulan
FROM mahasiswa;
```

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.01 seconds

	ID	NAMA	TANGGAL_LAHIR	HARI_TERAKHIR_BULAN
1	103122400001	Ahmad Wijaya	15-01-2004	31-01-2004
2	103122400002	Budi Santoso	22-03-2004	31-03-2004
3	103122400003	Citra Dewi	10-05-2004	31-05-2004
4	103122400004	Doni Prasetyo	18-07-2004	31-07-2004
5	103122400005	Nadia Tambunan	06-02-2004	29-02-2004
6	103122400006	Eka Putra	12-09-2004	30-09-2004
7	103122400007	Fiona Mayang	28-11-2004	30-11-2004
8	103122400008	Gina Septiani	05-04-2004	30-04-2004
9	103122400009	Hendra Gunawan	14-06-2004	30-06-2004
10	103122400010	Ica Ramadhani	21-08-2004	31-08-2004
11	103122400011	Jaka Permana	30-10-2004	31-10-2004
12	103122400012	Karina Sari	11-12-2004	31-12-2004
13	103122400013	Luki Hermawan	25-02-2004	29-02-2004
14	103122400014	Maya Kusuma	17-04-2004	30-04-2004
15	103122400015	Novita Salam	03-06-2004	30-06-2004

```sql
-- NEXT_DAY: Hari berikutnya dengan nama hari tertentu
SELECT
    id,
    nama,
    tanggal_lahir,
    NEXT_DAY(tanggal_lahir, 'MONDAY') AS next_monday
FROM mahasiswa;
```

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.012 seconds

	ID	NAMA	TANGGAL_LAHIR	NEXT_MONDAY
1	103122400001	Ahmad Wijaya	15-01-2004	19-01-2004
2	103122400002	Budi Santoso	22-03-2004	29-03-2004
3	103122400003	Citra Dewi	10-05-2004	17-05-2004
4	103122400004	Doni Prasetyo	18-07-2004	19-07-2004
5	103122400005	Nadia Tambunan	06-02-2004	09-02-2004
6	103122400006	Eka Putra	12-09-2004	13-09-2004
7	103122400007	Fiona Mayang	28-11-2004	29-11-2004
8	103122400008	Gina Septiani	05-04-2004	12-04-2004
9	103122400009	Hendra Gunawan	14-06-2004	21-06-2004
10	103122400010	Ica Ramadhani	21-08-2004	23-08-2004
11	103122400011	Jaka Permana	30-10-2004	01-11-2004
12	103122400012	Karina Sari	11-12-2004	13-12-2004
13	103122400013	Luki Hermawan	25-02-2004	01-03-2004
14	103122400014	Maya Kusuma	17-04-2004	19-04-2004
15	103122400015	Novita Salam	03-06-2004	07-06-2004

```sql
-- MONTHS_BETWEEN: Menghitung selisih bulan antara dua tanggal
SELECT
    id,
    nama,
    tanggal_lahir,
    ROUND(MONTHS_BETWEEN(SYSDATE, tanggal_lahir)) AS umur_bulan
FROM mahasiswa;
```

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.008 seconds

	ID	NAMA	TANGGAL_LAHIR	UMUR_BULAN
1	103122400001	Ahmad Wijaya	15-01-2004	266
2	103122400002	Budi Santoso	22-03-2004	263
3	103122400003	Citra Dewi	10-05-2004	262
4	103122400004	Doni Prasetyo	18-07-2004	260
5	103122400005	Nadia Tambunan	06-02-2004	265
6	103122400006	Eka Putra	12-09-2004	258
7	103122400007	Fiona Mayang	28-11-2004	255
8	103122400008	Gina Septiani	05-04-2004	263
9	103122400009	Hendra Gunawan	14-06-2004	261
10	103122400010	Ica Ramadhani	21-08-2004	258
11	103122400011	Jaka Permana	30-10-2004	256
12	103122400012	Karina Sari	11-12-2004	255
13	103122400013	Luki Hermawan	25-02-2004	264
14	103122400014	Maya Kusuma	17-04-2004	263
15	103122400015	Novita Salam	03-06-2004	261

```sql
-- ROUND dengan MONTH dan YEAR
SELECT
    id,
    nama,
    tanggal_lahir,
    ROUND(tanggal_lahir, 'MONTH') AS dibulatkan_ke_bulan,
    ROUND(tanggal_lahir, 'YEAR') AS dibulatkan_ke_tahun
FROM mahasiswa;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 15 in 0.011 seconds

	ID	NAMA	TANGGAL_LAHIR	DIBULATKAN_KE_BULAN	DIBULATKAN_KE_TAHUN
1	103122400001	Ahmad Wijaya	15-01-2004	01-01-2004	01-01-2004
2	103122400002	Budi Santoso	22-03-2004	01-04-2004	01-01-2004
3	103122400003	Citra Dewi	10-05-2004	01-05-2004	01-01-2004
4	103122400004	Doni Prasetyo	18-07-2004	01-08-2004	01-01-2005
5	103122400005	Nadia Tambunan	06-02-2004	01-02-2004	01-01-2004
6	103122400006	Eka Putra	12-09-2004	01-09-2004	01-01-2005
7	103122400007	Fiona Mayang	28-11-2004	01-12-2004	01-01-2005
8	103122400008	Gina Septiani	05-04-2004	01-04-2004	01-01-2004
9	103122400009	Hendra Gunawan	14-06-2004	01-06-2004	01-01-2004
10	103122400010	Ica Ramadhani	21-08-2004	01-09-2004	01-01-2005
11	103122400011	Jaka Permana	30-10-2004	01-11-2004	01-01-2005
12	103122400012	Karina Sari	11-12-2004	01-12-2004	01-01-2005
13	103122400013	Luki Hermawan	25-02-2004	01-03-2004	01-01-2004
14	103122400014	Maya Kusuma	17-04-2004	01-05-2004	01-01-2004
15	103122400015	Novita Salam	03-06-2004	01-06-2004	01-01-2004

c. Fungsi Konversi (TO_NUMBER (char), TO_CHAR (date), TO_CHAR (number), TO_DATE (char))

```sql
-- TO_NUMBER: Konversi string ke angka
SELECT
    id,
    TO_NUMBER('1234') AS angka_dari_string,
    TO_NUMBER('1234') + 100 AS hasil_operasi
FROM mahasiswa
WHERE ROWNUM <= 3;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 3 in 0.005 seconds

	ID	ANGKA_DARI_STRING	HASIL_OPERASI
1	103122400001	1234	1334
2	103122400002	1234	1334
3	103122400003	1234	1334

```sql
-- TO_CHAR untuk date: Konversi tanggal ke string dengan format
SELECT
    id,
    nama,
    tanggal_lahir,
    TO_CHAR(tanggal_lahir, 'DD/MM/YYYY') AS tanggal_format_1,
    TO_CHAR(tanggal_lahir, 'DD-MON-YYYY') AS tanggal_format_2
FROM mahasiswa;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 15 in 0.005 seconds

	ID	NAMA	TANGGAL_LAHIR	TANGGAL_FORMAT_1	TANGGAL_FORMAT_2
1	103122400001	Ahmad Wijaya	15-01-2004	15/01/2004	15-JAN-2004
2	103122400002	Budi Santoso	22-03-2004	22/03/2004	22-MAR-2004
3	103122400003	Citra Dewi	10-05-2004	10/05/2004	10-MAY-2004
4	103122400004	Doni Prasetyo	18-07-2004	18/07/2004	18-JUL-2004
5	103122400005	Nadia Tambunan	06-02-2004	06/02/2004	06-FEB-2004
6	103122400006	Eka Putra	12-09-2004	12/09/2004	12-SEP-2004
7	103122400007	Fiona Mayang	28-11-2004	28/11/2004	28-NOV-2004
8	103122400008	Gina Septiani	05-04-2004	05/04/2004	05-APR-2004
9	103122400009	Hendra Gunawan	14-06-2004	14/06/2004	14-JUN-2004
10	103122400010	Ica Ramadhani	21-08-2004	21/08/2004	21-AUG-2004
11	103122400011	Jaka Permana	30-10-2004	30/10/2004	30-OCT-2004
12	103122400012	Karina Sari	11-12-2004	11/12/2004	11-DEC-2004
13	103122400013	Luki Hermawan	25-02-2004	25/02/2004	25-FEB-2004
14	103122400014	Maya Kusuma	17-04-2004	17/04/2004	17-APR-2004
15	103122400015	Novita Salam	03-06-2004	03/06/2004	03-JUN-2004

```sql
-- TO_CHAR untuk number: Konversi angka ke string dengan format
SELECT
    id,
    tinggi_badan,
    berat_badan,
    TO_CHAR(tinggi_badan, '999') AS tinggi_format,
    TO_CHAR(berat_badan, '99.99') AS berat_format
FROM mahasiswa;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 15 in 0.004 seconds

	ID	TINGGI_BADAN	BERAT_BADAN	TINGGI_FORMAT	BERAT_FORMAT
1	103122400001	175	72	175	72.00
2	103122400002	172	68	172	68.00
3	103122400003	162	55	162	55.00
4	103122400004	178	75	178	75.00
5	103122400005	164	61	164	61.00
6	103122400006	170	70	170	70.00
7	103122400007	165	58	165	58.00
8	103122400008	160	52	160	52.00
9	103122400009	180	78	180	78.00
10	103122400010	163	56	163	56.00
11	103122400011	176	74	176	74.00
12	103122400012	164	60	164	60.00
13	103122400013	179	76	179	76.00
14	103122400014	161	54	161	54.00
15	103122400015	166	62	166	62.00

```sql
-- TO_DATE: Konversi string ke tanggal
SELECT
    id,
    nama,
    TO_DATE('25-12-2024', 'DD-MM-YYYY') AS tanggal_natal
FROM mahasiswa
WHERE ROWNUM <= 5;
```

Script Output | Query Result

SQL | All Rows Fetched: 5 in 0.004 seconds

	ID	NAMA	TANGGAL_NATAL
1	103122400001	Ahmad Wijaya	25-12-2024
2	103122400002	Budi Santoso	25-12-2024
3	103122400003	Citra Dewi	25-12-2024
4	103122400004	Doni Prasetyo	25-12-2024
5	103122400005	Nadia Tambunan	25-12-2024

d. Fungsi Numerik (ABS(n), MOD (m,n), FLOOR (n), POWER (m,n), ROUND (n[,m]), CEIL (n))

```sql
-- ABS: Nilai absolut
SELECT
    id,
    berat_badan,
    ABS(berat_badan - 70) AS selisih_dari_70
FROM mahasiswa;
```

cript Output | Query Result

SQL | All Rows Fetched: 15 in 0.004 seconds

	ID	BERAT_BADAN	SELISIH_DARI_70
1	103122400001	72	2
2	103122400002	68	2
3	103122400003	55	15
4	103122400004	75	5
5	103122400005	61	9
6	103122400006	70	0
7	103122400007	58	12
8	103122400008	52	18
9	103122400009	78	8
10	103122400010	56	14
11	103122400011	74	4
12	103122400012	60	10
13	103122400013	76	6
14	103122400014	54	16
15	103122400015	62	8

```sql
-- MOD: Sisa pembagian
SELECT
    id,
    berat_badan,
    MOD(berat_badan, 5) AS berat_mod_5
FROM mahasiswa;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 15 in 0.004 seconds

	ID	BERAT_BADAN	BERAT_MOD_5
1	103122400001	72	2
2	103122400002	68	3
3	103122400003	55	0
4	103122400004	75	0
5	103122400005	61	1
6	103122400006	70	0
7	103122400007	58	3
8	103122400008	52	2
9	103122400009	78	3
10	103122400010	56	1
11	103122400011	74	4
12	103122400012	60	0
13	103122400013	76	1
14	103122400014	54	4
15	103122400015	62	2

```sql
-- FLOOR: Pembulatan ke bawah
SELECT
    id,
    tinggi_badan,
    berat_badan,
    FLOOR(tinggi_badan / 10) * 10 AS tinggi_pembulatan_bawah
FROM mahasiswa;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 15 in 0.005 seconds

	ID	TINGGI_BADAN	BERAT_BADAN	TINGGI_PEMBULATAN_BAWAH
1	103122400001	175	72	170
2	103122400002	172	68	170
3	103122400003	162	55	160
4	103122400004	178	75	170
5	103122400005	164	61	160
6	103122400006	170	70	170
7	103122400007	165	58	160
8	103122400008	160	52	160
9	103122400009	180	78	180
10	103122400010	163	56	160
11	103122400011	176	74	170
12	103122400012	164	60	160
13	103122400013	179	76	170
14	103122400014	161	54	160
15	103122400015	166	62	160

```sql
-- POWER: Pangkat
SELECT
    id,
    berat_badan,
    POWER(berat_badan, 2) AS berat_kuadrat,
    POWER(berat_badan, 0.5) AS akar_berat
FROM mahasiswa;
```

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.006 seconds

	ID	BERAT_BADAN	BERAT_KUADRAT	AKAR_BERAT
1	103122400001	72	5184	8,4852813742385702928101323452581884714 2
2	103122400002	68	4624	8,2462112512353210996428197119481540503 5
3	103122400003	55	3025	7,4161984870956629487113974408007130608 8
4	103122400004	75	5625	8,6602540378443864676372317075293618347 4
5	103122400005	61	3721	7,8102496759066543941297227357591014135 4
6	103122400006	70	4900	8,3666002653407554797817202578518748939 3
7	103122400007	58	3364	7,6157731058639082856614110271583230054 1
8	103122400008	52	2704	7,2111025509279785862384425349409918925 5
9	103122400009	78	6084	8,8317608663278468547640427269592539642 5
10	103122400010	56	3136	7,4833147735478827711674974646330986034 9
11	103122400011	74	5476	8,6023252670426267717294735350497136320 8
12	103122400012	60	3600	7,7459666924148337703585307995647992216 4
13	103122400013	76	5776	8,7177978870813471044739639677192313183 4
14	103122400014	54	2916	7,3484692283495342945918522241176741759 1
15	103122400015	62	3844	7,8740078740118110196850344488120078636 4

```sql
-- ROUND: Pembulatan
SELECT
    id,
    tinggi_badan,
    berat_badan,
    ROUND(tinggi_badan / 10) * 10 AS tinggi_rounded,
    ROUND(berat_badan / 5) * 5 AS berat_rounded_5
FROM mahasiswa;
```

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.004 seconds

	ID	TINGGI_BADAN	BERAT_BADAN	TINGGI_ROUNDED	BERAT_ROUNDED_5
1	103122400001	175	72	180	70
2	103122400002	172	68	170	70
3	103122400003	162	55	160	55
4	103122400004	178	75	180	75
5	103122400005	164	61	160	60
6	103122400006	170	70	170	70
7	103122400007	165	58	170	60
8	103122400008	160	52	160	50
9	103122400009	180	78	180	80
10	103122400010	163	56	160	55
11	103122400011	176	74	180	75
12	103122400012	164	60	160	60
13	103122400013	179	76	180	75
14	103122400014	161	54	160	55
15	103122400015	166	62	170	60

```sql
-- CEIL: Pembulatan ke atas
SELECT
    id,
    tinggi_badan,
    berat_badan,
    CEIL(tinggi_badan / 10) * 10 AS tinggi_pembulatan_atas
FROM mahasiswa;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 15 in 0.006 seconds

	ID	TINGGI_BADAN	BERAT_BADAN	TINGGI_PEMBULATAN_ATAS
1	103122400001	175	72	180
2	103122400002	172	68	180
3	103122400003	162	55	170
4	103122400004	178	75	180
5	103122400005	164	61	170
6	103122400006	170	70	170
7	103122400007	165	58	170
8	103122400008	160	52	160
9	103122400009	180	78	180
10	103122400010	163	56	170
11	103122400011	176	74	180
12	103122400012	164	60	170
13	103122400013	179	76	180
14	103122400014	161	54	170
15	103122400015	166	62	170